

Speakers



Abhay Patwardhan, Director,
EV Business, Chheda Electricals
and Electronics Pvt Ltd



Piyush Gupta, CEO,
Lithion Power



Sangeet Kumar, CEO,
Addverb



Sanjeev Keskar, CEO,
Arvind Consultancy



Supratim Naskar, Co-founder
and CTO, Cell Propulsion

placement techniques. For instance, a user may not want fast charging if he's charging at home.

Is fast charging a myth?

Patwardhan opines that fast charging in a true sense is a myth. "Because as you try to charge faster, it will charge to a low extent. So, there is going to be a trade-off. Second, you may end up increasing the temperature of the battery. So, how useful it turns out after that for the drive condition becomes a performance challenge."

From a user perspective, a fast charger of 20kW needs a 3-phase connection. Now, for day-to-day operations, we do not get this 3-phase connection of so much capacity available easily. Also, there is hardly any room for management of heat generated by the batteries. Patwardhan explains, "Maybe the two-wheeler battery packs are smaller and easy to swap. A big battery pack from a vehicle will need a big replacement mechanism. Only time will tell its feasibility."

Areas to build expertise

Naskar suggests stationary energy storage as a definite option to build expertise in terms of batteries to remove the pain point, as power back-ups are used in telecom industries, households, and more. The other area he points to is drone technology. He adds, "I think the batteries that we are using for a standard warehouse robot, we can repurpose and maybe use for a similar application for e-bikes. Not may be e-scooters, but electric powered bicycles."

Patwardhan says the first major challenge in a battery pack is the packaging because you will never get enough space to build what you want to, and then there is the issue of thermal management. These two are the backbone of constructing any battery pack meaningfully for any application.

EV ecosystem

EV manufacturing in India is mostly a cost-driven exercise. For instance, a Bouguereau price point is \$1,800

and it gives a range of 80km. With the same vehicle and performance expectations in India, no one would be willing to pay more than \$800.

Patwardhan sees this as an opportunity. He says, "With vehicle manufacturers pointing out that the battery represents 30 or 40% of the total cost, doing a battery business actually means I am doing little less than half of the vehicle business." He strongly believes that EVs would be sold without battery and battery fitment would eventually become universal, giving the user an option to decide whether he needs it for only 50km or 100km.

EV technology is going to be a big revolution and a great opportunity for India to take global leadership. There will be a requirement for motor control application, battery management system, charging station, DC-to-DC converters, and many more things related to electronics, which will drive the demand for semiconductors, connectors, and passive components. The innovation to suit Indian conditions will be the critical requirement as far as the electronic embedded part of the vehicle is concerned.

With India having a strong backbone in the MSME sector and manufacturing, Naskar says he does not find any limitation for battery manufacture, in general. By extending the boundaries from domestic to international, India would get a huge cost advantage in the international markets. He adds, "I think it's a good opportunity and a lot of investment push is coming from the government as well. We are completely set to get into this revolution and take the lead." **EFY**

The article was prepared by Abhinaya Kuzhanthaivel, an Assistant Editor at EFY, based on the panel discussion held during Battery Tech Show 2021 among Piyush Gupta, CEO, Lithion Power; Supratim Naskar, Co-founder and CTO, Cell Propulsion; Sangeet Kumar, CEO, Addverb; Sanjeev Keskar, CEO, Arvind Consultancy; and Abhay Patwardhan, Director, EV Business, Chheda Electricals And Electronics Pvt Ltd.